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Q. I am an engineering student. While learning the subject of very-large-scale integration (VLSI), I could not understand the following terms:

1. Setup time and hold time
2. Clock skew

Kindly explain.

Neha Sharma

Sonipat

A. These terms related to the study of metastability factor of VLSI design are explained below:

Setup time. It is related to the stability of clock pulse. It refers to the amount of time before the clock edge that the input signal needs to be stable in order to be accepted properly on the clock edge. The clock pulse becomes functional after setup time.

Hold time. It is the amount of time for which the input signal remains valid after the clock edge. It ensures that the signal is sensed properly at the clock edge after setup time.

Whenever there are violations of setup and hold time in any flip-flop, it enters a state where its output is unpredictable. This state is known as metastable state (quasi stable state). At the end of this state, the flip-flop settles down to either '1' or '0.' This whole process is known as metastability.

Clock skew. Clock skew is a phenomenon in synchronous circuits in which the clock signal (sent from the clock circuit) arrives at different components at different times.

This is typically due to two factors:

Material flaw. It causes a signal to

travel faster or slower than expected.

Distance. If the signal has to travel the entire length of a circuit, it is likely to arrive at different parts of the circuit at different times depending on the circuit's size.

Clock skew can harm in two ways:

1. Suppose that a logic path travels through combinational logic from a source flip-flop to a destination flip-flop. If the destination flip-flop receives the clock tick later than the source flip-flop, and if the logic path delay is short enough, the data signal might arrive at the destination flip-flop before the clock tick, destroying the previous data that should have been clocked through. This is called a hold violation because the previous data is not held long enough at the destination flip-flop to be properly clocked through.

2. If the destination flip-flop receives the clock tick earlier than the source flip-flop, the data signal has that much less time to reach the destination flip-flop before the next clock tick. If it fails to do so, a setup violation occurs. It is so-called because the new data is not set up and stable before the next clock tick arrived.

A hold violation is more serious than a setup violation because it cannot be fixed by increasing the clock period. Clock skew, if done right, can also benefit a circuit. It can be intentionally introduced to decrease the clock period at which the circuit will operate correctly and/or to increase the setup or hold safety margins.

The optimal set of clock delays is determined by a linear program in which a setup and a hold constraint appear for each logic path. In this linear program, zero clock skew is merely a feasible point. Clock skew can be minimised by proper routing of the clock signal (clock distribution tree) or putting a variable delay buffer so that all clock inputs arrive at the same time.

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